

## DAY 9

| Aspect      | Fact table                                                                          | Dimension table                                                                     |
|-------------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Definition  | Stores <b>**quantitative data (measures/facts)**</b> about the business.            | Stores <b>**descriptive data (attributes)**</b> that describe dimensions of facts.  |
| Content     | Contains numbers: sales amount, quantity, profit, cost, etc.                        | Contains text/labels: product name, customer name, location, time, etc.             |
| Granularity | Very detailed & large (millions of rows).                                           | Comparatively Smaller                                                               |
| Primary Key | Composite key (foreign keys referencing dimension tables).                          | Single primary key (referenced in fact table).                                      |
| Nature      | Dynamic → grows continuously as transactions happen.                                | Static → changes less frequently (e.g., product details, customer info).            |
| Example     | `SalesFact` (SaleID, ProductID, CustomerID, DateID, Amount, Quantity)`              | `ProductDim` (ProductID, ProductName, Category, Price)`                             |
| Use         | For <b>**calculations &amp; analytics**</b> (e.g., total sales, profit per region). | For <b>**filtering, grouping, slicing**</b> (e.g., sales by category, region, year) |

### (A) IF EXISTS

Used to check if something exists before performing an action.

-- Drop a table only if it exists

IF EXISTS (SELECT \* FROM sys.objects WHERE name = 'SalesFact' AND type = 'U')

DROP TABLE SalesFact;

### (B) LIKE

Used for pattern matching in strings.

-- Find all employees whose name starts with 'A'

SELECT \*

FROM Employees

WHERE FirstName LIKE 'A%';

-- Find customers whose email ends with 'gmail.com'

SELECT \*

```
FROM Customers  
WHERE Email LIKE '%@gmail.com';
```

## **(C) EXISTS**

Checks if a subquery returns any rows (boolean test).

-- Find employees who have at least one order

```
SELECT e.EmployeeID, e.FirstName, e.LastName  
FROM Employees e  
WHERE EXISTS (  
    SELECT 1  
    FROM Orders o  
    WHERE o.EmployeeID = e.EmployeeID  
);
```

## **(D) VIEW**

A stored query that acts like a virtual table.

-- Create a view for Sales with Customer details

```
CREATE VIEW SalesWithCustomer AS  
SELECT s.SaleID, s.Amount, s.Quantity, c.CustomerName, c.City  
FROM SalesFact s  
JOIN CustomerDim c  
    ON s.CustomerID = c.CustomerID;
```

**-- Use the view**

```
SELECT * FROM SalesWithCustomer WHERE City = 'Delhi';
```